

Curl of a Vector

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13- Curl of a Vector Field

13.1 We have already see two uses of $\vec{\nabla}$ operator, namely

$$\text{gradient of a scalar field: } \vec{\nabla} f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$$

$$\text{divergence of a vector field: } \vec{\nabla} \cdot \vec{v} = \frac{\partial}{\partial x} v_1 + \frac{\partial}{\partial y} v_2 + \frac{\partial}{\partial z} v_3$$

Note that these are not limited to $\mathbb{R}^3$.

$$\text{In } \mathbb{R}^2: \quad \vec{\nabla} f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}, \quad \vec{\nabla} \cdot \vec{v} = \frac{\partial}{\partial x} v_1 + \frac{\partial}{\partial y} v_2$$

In $\mathbb{R}^4$ (coordinates w, x, y, z):

$$\vec{\nabla} f = \left[\frac{\partial f}{\partial w}, \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right]$$

$$\vec{\nabla} \cdot \vec{v} = \frac{\partial}{\partial w} v_1 + \frac{\partial}{\partial x} v_2 + \frac{\partial}{\partial y} v_3 + \frac{\partial}{\partial z} v_4$$

Today we will see another use of $\vec{\nabla}$ in computing the curl of a vector field

13.2

Let $\mathbf{v}(x, y, z) = [v_1, v_2, v_3] = v_1 \mathbf{i} + v_2 \mathbf{j} + v_3 \mathbf{k}$ be a differentiable vector function of the Cartesian coordinates x, y, z . Then the **curl** of the vector function $\mathbf{v}$ or of the vector field given by $\mathbf{v}$ is defined by the "symbolic" determinant

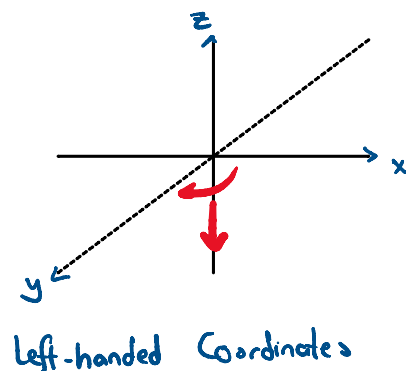
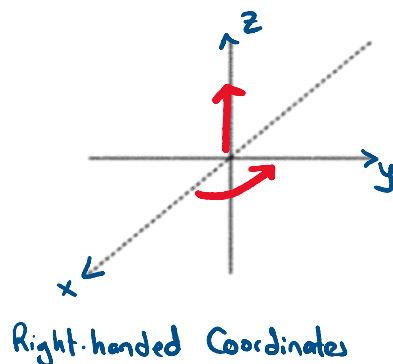
$$\begin{aligned} (1) \quad \text{curl } \mathbf{v} = \nabla \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} \\ &= \left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial v_1}{\partial z} - \frac{\partial v_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v_2}{\partial x} - \frac{\partial v_1}{\partial y} \right) \mathbf{k}. \end{aligned}$$

13.3 Notes:

(1) The "cross-product" is only defined in $\mathbb{R}^3$, so technically we can only take the curl of a vector with exactly three components $\vec{v}_1, \vec{v}_2, \vec{v}_3$ that depend on three-space: x, y, z .

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(2) Cross-products follow a right-hand rule; in particular the coordinates have to be right-handed (see picture).



13.4 Example: $\vec{v} = [e^{-z^2}, e^{-x^2}, e^{-y^2}]$

$$\begin{aligned}\vec{\nabla} \times \vec{v} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{-z^2} & e^{-x^2} & e^{-y^2} \end{vmatrix} = \hat{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{-x^2} & e^{-y^2} \end{vmatrix} - \hat{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ e^{-z^2} & e^{-y^2} \end{vmatrix} + \hat{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ e^{-z^2} & e^{-x^2} \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial}{\partial y} e^{-y^2} - \frac{\partial}{\partial z} e^{-x^2} \right) - \hat{j} \left(\frac{\partial}{\partial x} e^{-y^2} - \frac{\partial}{\partial z} e^{-z^2} \right) + \hat{k} \left(\frac{\partial}{\partial x} e^{-x^2} - \frac{\partial}{\partial y} e^{-z^2} \right) \\ &= \hat{i} \left[(-2y)e^{-y^2} - 0 \right] - \hat{j} \left[0 - (-2z)e^{-z^2} \right] + \hat{k} \left[(-2x)e^{-x^2} - 0 \right] \\ &= -2ye^{-y^2}\hat{i} - 2ze^{-z^2}\hat{j} - 2xe^{-x^2}\hat{k} \\ &= -2 \begin{bmatrix} ye^{-y^2} \\ ze^{-z^2} \\ xe^{-x^2} \end{bmatrix}\end{aligned}$$

13.5 Example: $\vec{v} = \frac{1}{(x^2+y^2+z^2)^{3/2}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\vec{\nabla} \times \vec{v} = \left[\frac{\partial}{\partial y} \frac{1}{(x^2+y^2+z^2)^{3/2}} z - \frac{\partial}{\partial z} \frac{1}{(x^2+y^2+z^2)^{3/2}} y \right] \hat{i}$$

$$\begin{aligned}
& \left[\frac{\partial}{\partial y} \frac{1}{(x^2+y^2+z^2)^{3/2}} - \frac{\partial}{\partial z} \frac{1}{(x^2+y^2+z^2)^{3/2}} \right] \hat{i} \\
& + \left[\frac{\partial}{\partial z} \frac{1}{(x^2+y^2+z^2)^{3/2}} - \frac{\partial}{\partial x} \frac{1}{(x^2+y^2+z^2)^{3/2}} \right] \hat{j} \\
& + \left[\frac{\partial}{\partial x} \frac{1}{(x^2+y^2+z^2)^{3/2}} - \frac{\partial}{\partial y} \frac{1}{(x^2+y^2+z^2)^{3/2}} \right] \hat{k} \\
\Rightarrow \vec{\nabla} \times \vec{v} &= \left[z \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2y) - y \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2z) \right] \hat{i} \\
& + \left[x \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2z) - z \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2x) \right] \hat{j} \\
& + \left[y \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2x) - x \left(-\frac{3}{2} \right) (x^2+y^2+z^2)^{-5/2} (2y) \right] \hat{k} \\
\Rightarrow \vec{\nabla} \times \vec{v} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \vec{0}
\end{aligned}$$

13.6 Rotation of a Rigid Body. Relation to the Curl

We have seen in Example 5, Sec. 9.3, that a rotation of a rigid body B about a fixed axis in space can be described by a vector $\mathbf{w}$ of magnitude ω in the direction of the axis of rotation, where $\omega (>0)$ is the angular speed of the rotation, and $\mathbf{w}$ is directed so that the rotation appears clockwise if we look in the direction of $\mathbf{w}$. According to (9), Sec. 9.3, the velocity field of the rotation can be represented in the form

$$\mathbf{v} = \mathbf{w} \times \mathbf{r}$$

where $\mathbf{r}$ is the position vector of a moving point with respect to a Cartesian coordinate system *having the origin on the axis of rotation*. Let us choose right-handed Cartesian coordinates such that the axis of rotation is the z -axis. Then (see Example 2 in Sec. 9.4)

$$\mathbf{w} = [0, 0, \omega] = \omega \mathbf{k}, \quad \mathbf{v} = \mathbf{w} \times \mathbf{r} = [-\omega y, \omega x, 0] = -\omega y \mathbf{i} + \omega x \mathbf{j}.$$

Hence

$$\text{curl } \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\omega y & \omega x & 0 \end{vmatrix} = [0, 0, 2\omega] = 2\omega \mathbf{k} = 2\mathbf{w}.$$

13.7

THEOREM

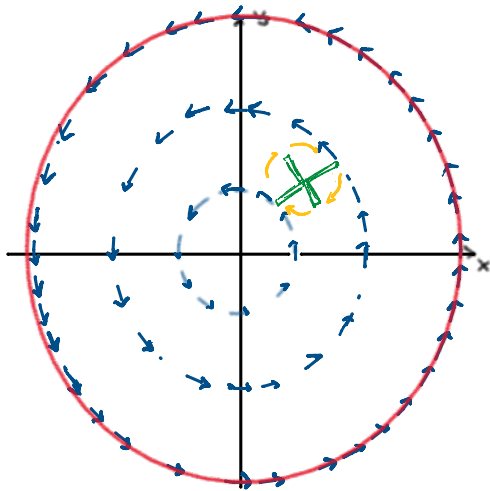
Rotating Body and Curl

The curl of the velocity field of a rotating rigid body has the direction of the axis of the rotation, and its magnitude equals twice the angular speed of the rotation.

Rotating Body and Curl

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12.8 Be careful! The rotation is local in the following sense:



$$\vec{v} = \frac{1}{\sqrt{x^2+y^2}} (-y\hat{i} + x\hat{j} + 0\hat{k})$$

$$\vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{-y}{\sqrt{x^2+y^2}} & \frac{x}{\sqrt{x^2+y^2}} & 0 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} 0 - \frac{\partial}{\partial z} \frac{x}{\sqrt{x^2+y^2}} \right]$$

$$- \hat{j} \left[\frac{\partial}{\partial x} 0 - \frac{\partial}{\partial z} \frac{-y}{\sqrt{x^2+y^2}} \right]$$

$$+ \hat{k} \left[\frac{\partial}{\partial x} \left(\frac{x}{\sqrt{x^2+y^2}} \right) - \frac{\partial}{\partial y} \left(\frac{-y}{\sqrt{x^2+y^2}} \right) \right]$$

$$= 0\hat{i} - 0\hat{j} + \hat{k} \left[\left(\frac{\sqrt{x^2+y^2}(1) - x(\frac{1}{2})(x^2+y^2)^{-1/2}(2x)}{x^2+y^2} \right) + \left(\frac{\sqrt{x^2+y^2}(1) - y(\frac{1}{2})(x^2+y^2)^{-1/2}(2y)}{x^2+y^2} \right) \right]$$

$$= 0\hat{i} + 0\hat{j} + \hat{k} \left[\frac{(x^2+y^2)^{1/2} - x^2(x^2+y^2)^{-1/2}}{(x^2+y^2)} + \frac{(x^2+y^2)^{1/2} - y^2(x^2+y^2)^{-1/2}}{(x^2+y^2)} \right]$$

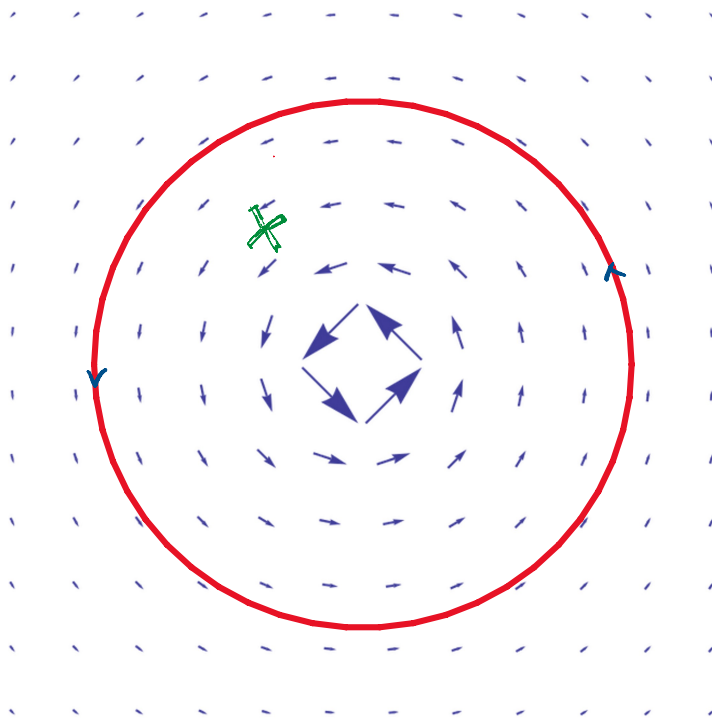
$$= 0\hat{i} + 0\hat{j} + \hat{k} \left[\frac{x^2+y^2 - x^2}{(x^2+y^2)^{3/2}} + \frac{x^2+y^2 - y^2}{(x^2+y^2)^{3/2}} \right] = \hat{k} \left[\frac{1}{\sqrt{x^2+y^2}} \right]$$

Slide: Find $\vec{\nabla} \times \left[\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0 \right]$.

On the other hand:

if $\vec{v} = \left(-\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0 \right)$, then $\vec{\nabla} \times \vec{v} = \vec{0}$

..... No local rotation, even though there is a global rotation.



No local rotation, even though there is a global rotation

Think of the global rotation more like a revolution around the center, and the local rotation similar to how the Earth rotates around its axis.

13.9 Notice that we have been sneaky and computed curl of 2D vectors by the technique of embedding:

$$\vec{v}(x, y) = \begin{bmatrix} v_1(x, y) \\ v_2(x, y) \end{bmatrix} \xrightarrow{\text{embedding}} \vec{v}(x, y, z) = \begin{bmatrix} v_1(x, y) \\ v_2(x, y) \\ 0 \end{bmatrix} \quad \text{so that} \quad \frac{\partial}{\partial z} v_1 = \frac{\partial}{\partial z} v_2 = 0$$

Then we compute

$$\vec{\nabla} \times \vec{v} = \hat{k} \left[\frac{\partial}{\partial x} v_2 - \frac{\partial}{\partial y} v_1 \right] = \begin{bmatrix} 0 \\ 0 \\ \frac{\partial}{\partial x} v_2 - \frac{\partial}{\partial y} v_1 \end{bmatrix}$$

Looks sus, but actually this has its use in Green's theorem (next week)

13.10

THEOREM 2

Grad, Div, Curl

Gradient fields are **irrotational**. That is, if a continuously differentiable vector function is the gradient of a scalar function f , then its curl is the zero vector,

(2)

$$\text{curl}(\text{grad } f) = \mathbf{0}.$$

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$$(2) \quad \text{curl}(\text{grad } f) = \mathbf{0}.$$

Furthermore, the divergence of the curl of a twice continuously differentiable vector function $\mathbf{v}$ is zero,

$$(3) \quad \text{div}(\text{curl } \mathbf{v}) = 0.$$

13.11 In general:

If $\vec{F} = \vec{\nabla} f$, we say f is a potential
 $\vec{F}$ is a conservative field

so then $\int_C \vec{F} \cdot d\vec{r}$ is path-independent, i.e. $\int_A^B \vec{F}$ does not depend on the curve taken from A to B.

and also $\vec{\nabla} \times \vec{F} = \vec{0}$, so $\vec{F}$ is irrotational

13.12

THEOREM 3

Invariance of the Curl

$\text{curl } \mathbf{v}$ is a vector. It has a length and a direction that are independent of the particular choice of a Cartesian coordinate system in space.

13.13 There is nothing special about Cartesian Coordinates; in any right-handed coordinate, $\text{curl}(\vec{v})$ has the same length and direction.

13.14 Important Properties of Curl:

- (a) $\text{curl}(\mathbf{u} + \mathbf{v}) = \text{curl } \mathbf{u} + \text{curl } \mathbf{v}$
- (b) $\text{div}(\text{curl } \mathbf{v}) = 0$
- (c) $\text{curl}(f\mathbf{v}) = (\text{grad } f) \times \mathbf{v} + f \text{curl } \mathbf{v}$
- (d) $\text{curl}(\text{grad } f) = \mathbf{0}$
- (e) $\text{div}(\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot \text{curl } \mathbf{u} - \mathbf{u} \cdot \text{curl } \mathbf{v}$